Object-Conditioned Strategy-Level Joint Parameter-Bundle Configuration for Haptic Teleoperation

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Abstract

Fixed teleoperation settings are difficult to tune across objects with different fragility, geometry, and grasping requirements. Existing object-conditioned methods primarily adapt slave-side impedance, whereas coordinated configuration of slave-side impedance, operator-facing haptic feedback, and gripper execution remains underexplored. This paper proposes an object-conditioned strategy-layer framework that maps each detected object class to one of three operational strategies and jointly assigns parameter bundles to these subsystems. Low-rate visual perception is asynchronously integrated with a 200-Hz supervisory loop. The first confidence-qualified detection triggers a one-shot configuration, while strategy locking prevents repeated switching within a trial. Three operators completed 27 matched task blocks, yielding 135 grasp-and-transfer trials across six objects and five control modes. Compared with impedance-only reconfiguration, the joint parameter-bundle mode reduced mean completion time by 1.79 s (8.5%; matched-block bootstrap 95% CI, 1.10–2.51 s). The reduction was directionally consistent across all operators and objects. Descriptively, this mode also achieved the highest observed success rate and the lowest Raw NASA-TLX score. These findings indicate a system-level advantage of bundled configuration under the tested conditions, but do not isolate the effects of individual channels. Overall, the framework provides an interpretable, one-shot configuration approach for heterogeneous-object haptic teleoperation.

Keywords: haptic teleoperation, impedance control, telemanipulation, object-conditioned parameter configuration, strategy-level joint parameter-bundle configuration, human-in-the-loop evaluation

1. Introduction

Teleoperation brings human judgment to remote manipulation tasks that remain difficult for autonomous robots, including hazardous operation, flexible manufacturing, and unstructured manipulation. Haptic feedback can convey remote contact information to the operator and support assessment of contact state, grasp stability, and slip. Consequently, a practical teleoperation system must integrate bilateral control with perception, mechanical execution, and the human–machine interface [1, 2].

For heterogeneous objects, one fixed setting cannot simultaneously provide gentle contact and robust positioning, grasping, and transport. Lower stiffness and grasp force can reduce contact severity but can impair positioning and holding stability; higher settings can improve positional robustness while increasing the risk of impact or deformation. This trade-off motivates object-conditioned configuration across the subsystems involved in telemanipulation, rather than a single fixed setting for all objects.

Impedance control regulates the relationship among displacement, velocity, and interaction force and is a standard means of achieving compliant robot behavior [3]. Variable-impedance methods adapt stiffness and damping to task state, contact information, learned policies, or human state, while addressing the stability implications of time-varying parameters [4, 5]. In teleoperation, tele-impedance and bilateral-control studies have regulated remote-robot stiffness according to operator motion, estimated human impedance, contact state, or task phase [6–12]. Related work has also examined operator-facing haptic feedback and tactile feedback for telemanipulation [13–15]. These studies establish important approaches to remote mechanical compliance and haptic interaction, but their principal focus is commonly on impedance transmission, stiffness regulation, transparency, or feedback presentation.

Visual and multimodal information has extended semi-autonomous tele-impedance beyond contact-driven adaptation. Vision and voice have been used to select slave-side impedance from object properties with operator confirmation or correction [16]; visual-force control has combined visual and force information in a visual-feature space [17]; and task-oriented shared control has reduced manual decision-making [18]. More recent work has derived stiffness from object geometry, material, and environmental relationships [19], or generated task-relevant stiffness matrices from visual, gaze, and language inputs [20]. Automatic switching and passive arbitration address when control authority or force/stiffness scaling should change [21, 22], while teleoperated grasping has used tactile deformation and grip-safety information to regulate gripper impedance [23]. Thus, object- and task-dependent parameter adaptation is established; in particular, visual information can be used to select or generate remote impedance settings.

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The remaining system-level question is not whether vision can adjust impedance, but whether one interpretable object-conditioned strategy can jointly configure the remote arm, the operator-facing haptic interface, and gripper execution. In the closely related literature considered here, these elements are usually adjusted as separate design components or the object information is translated primarily into remote stiffness, impedance matrices, or control-authority decisions. We therefore investigate whether a common parameter bundle spanning these three subsystems provides a task-level benefit beyond visual-semantic display, manual strategy selection, or impedance-only reconfiguration.

Joint configuration also creates an integration problem: visual inference is intermittent and can be uncertain, whereas the teleoperation update must remain responsive to the operator. Passing every perception result directly into the control path can either delay the supervisory update or induce repeated switching when detections fluctuate. The required architecture must therefore transfer semantic information into a limited and consistent configuration action without turning the system into continuous visual servoing.

To address this question, we propose an object-conditioned joint-configuration framework for haptic teleoperation of heterogeneous objects. Each detected object class is mapped to one of three operational strategies and then to a parameter bundle specifying slave-side impedance, master-side haptic rendering, and gripper execution. After task initiation, the first threshold-qualified mapped detection can initiate a one-shot configuration event, after which the selected strategy is locked for the remainder of the trial. A multi-rate implementation keeps visual inference outside the supervisory teleoperation call path and prevents repeated reconfiguration caused by fluctuating detections. The framework is evaluated on a closed set of six known object instances using five modes: a fixed-parameter baseline (A), a manual-selection workflow baseline (B), joint parameter-bundle configuration (C), visual-semantic display with fixed parameters (D), and impedance-only reconfiguration (E). The primary C–E comparison tests the system-level contribution of the complete bundle under matched visual-semantic and impedance conditions, whereas C–D tests the contribution beyond the displayed visual-semantic information.

The main contributions are as follows:

1. **Object-conditioned cross-subsystem configuration.** An interpretable strategy layer maps each detected object class to an operational strategy and jointly assigns a parameter bundle across slave-side impedance, master-side haptic rendering, and gripper execution.
2. **Multi-rate one-shot integration.** A non-blocking architecture decouples low-rate visual inference from the teleoperation update and converts a qualified semantic result into a bounded, one-shot configuration event with intra-trial strategy locking.
3. **Matched human-in-the-loop evaluation.** A five-mode experiment on an Omega.7–Panda platform, involving three operators, six heterogeneous objects, and 135 trials, evaluates whether joint configuration provides an additional system-level benefit over visual-semantic display and impedance-only reconfiguration.

The evaluation is designed to assess the bundled configuration at the system level. It does not identify the independent causal contributions of the haptic-interface and gripper parameter groups, and its empirical scope is limited to the six known objects and repeated measurements in the present sample.

2. Methods and System Implementation

2.1. System Overview and Multi-Rate Architecture

The experimental platform comprised an Omega.7 force-feedback master device, a seven-degree-of-freedom Franka Panda robot, a Franka Hand gripper [24], an Intel RealSense D435i camera, and a control computer (Fig. 1). The D435i provided 424 × 240-pixel color images at 15 fps; the depth stream was not used. The control computer ran visual detection, the nominal 200-Hz supervisory teleoperation update, parameter configuration, haptic rendering, gripper control, and data logging.

Figure 2 illustrates the timing relationships within this multi-rate asynchronous software architecture.

RGB frames were acquired every 66.7 ms and written to a single-slot queue, preventing stale images from accumulating. YOLO11n ran in an independent process and returned class labels and confidence scores through a two-slot result queue. The nominal 200-Hz supervisory loop polled this queue every 5 ms without waiting for inference, allowing camera acquisition, perception, and operator motion to proceed in parallel. Per-image processing time was characterized separately in a controlled image test. Table 1 summarizes the update timing, triggers, and data flow of the system modules.

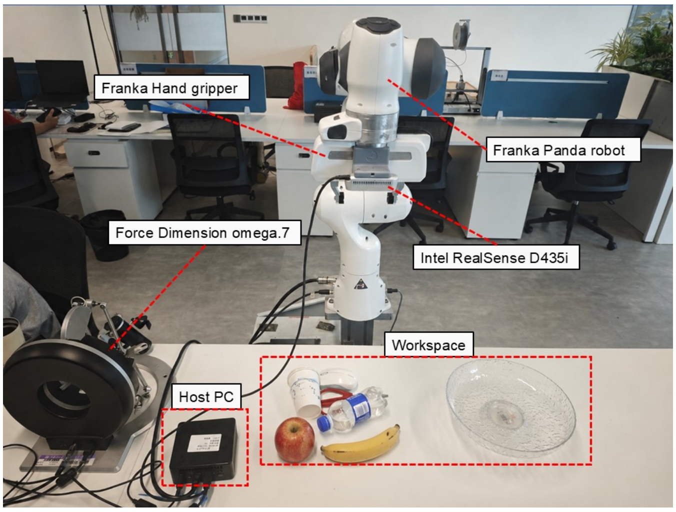

Figure 1: Experimental teleoperation platform showing the Omega.7 master device, Franka Panda robot, Intel RealSense D435i camera, host computer, and object workspace. The Franka Hand gripper is mounted at the Panda end effector.

Table 1: Runtime characteristics and data flow of the multi-rate mechatronic system.

Module	Timing or trigger	Data flow
Master input	200 Hz	Omega.7 translational position and buttons → master-state update.
Slave control	200 Hz	Desired pose and impedance parameters → Panda command.
Image acquisition	15 fps (66.7 ms/frame)	D435i color stream → 424 × 240 color image.
YOLO11n inference	Asynchronous; controlled-test performance characterized separately	Latest available image → class and confidence.
Strategy scheduling	First threshold-qualified detection	Class and confidence → target $\Theta(c)$.
Haptic rendering	200 Hz	Force-related state → Omega.7 force command.
Gripper command	Button event	Gripper input and command parameters → Franka Hand command.

Note: YOLO11n inference is executed in an independent process and does not block the supervisory loop.

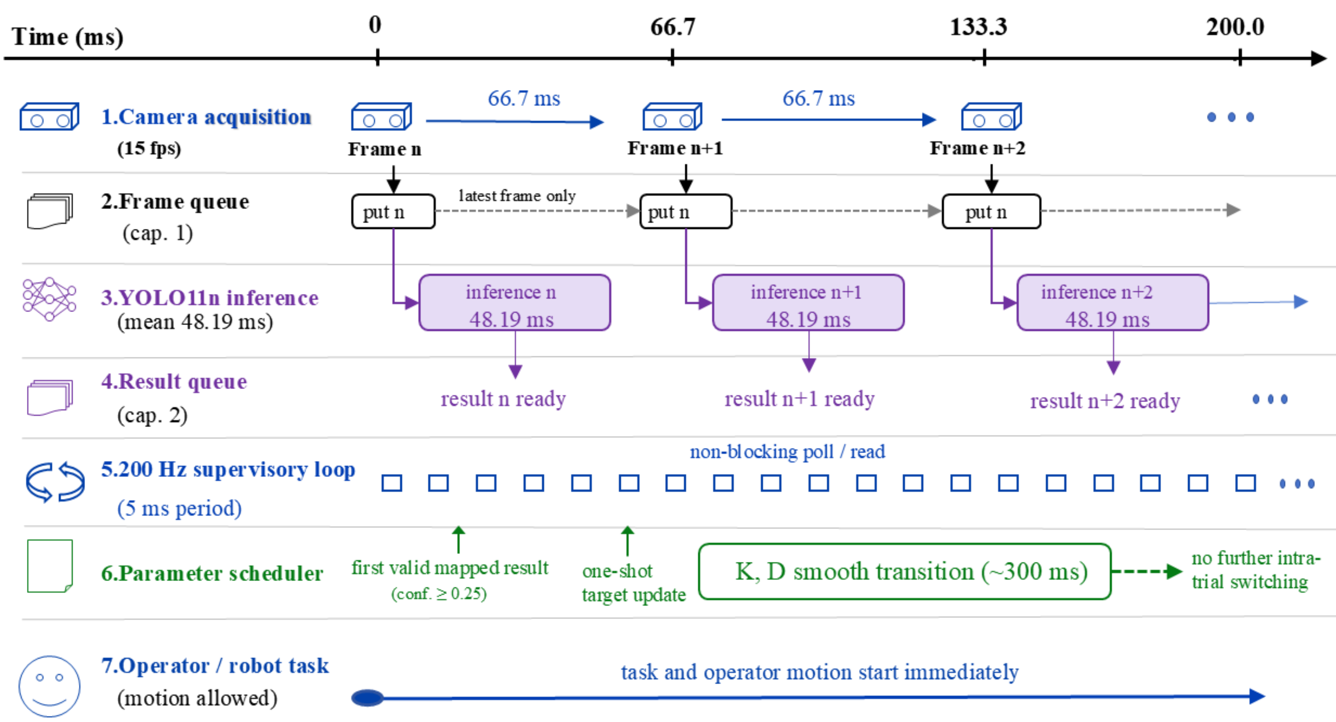

Figure 2: Schematic, not-to-scale timing of the asynchronous perception–control implementation. Frames and results are exchanged through bounded queues while the nominal 200-Hz supervisory loop polls without blocking. The first valid mapped result (≥ 0.25 confidence) initiates one locked update; impedance parameters transition over approximately 300 ms. This schematic shows nominal timing, not measured end-to-end latency.

2.2. *Teleoperation and Mechatronic Control Channels*

2.2.1. *Master–slave motion mapping*

Only translational motion of the Omega.7 was mapped to the Panda. The master-device position increment and desired slave position were updated as

$$\Delta \mathbf{p}_m(k) = \mathbf{p}_m(k) - \mathbf{p}_m(k-1), \tag{1}$$

$$\mathbf{p}_d(k) = \mathbf{p}_d(k-1) + S\,\mathbf{C}\Delta \mathbf{p}_m(k), \tag{2}$$

where the position-scaling factor was $S = 3.0$ and $\mathbf{C} = \mathrm{diag}(-1, -1, 1)$. The desired end-effector orientation $\mathbf{R}_d$ was fixed at its prescribed value throughout each trial; active rotational input from the operator was not mapped.

2.2.2. *Slave-side Cartesian impedance*

At a conceptual level, Cartesian impedance control acted on the six-dimensional pose error,

$$\mathbf{w}_c = \mathbf{K}(c)\mathbf{e} + \mathbf{D}(c)\dot{\mathbf{e}}, \tag{3}$$

$$\mathbf{K}(c) = \mathrm{diag}(K_t, K_t, K_t, K_r, K_r, K_r), \tag{4}$$

where $\mathbf{e}$ contains translational error relative to $\mathbf{p}_d$ and rotational error relative to the fixed $\mathbf{R}_d$. Thus, K_r changes resistance to rotational disturbances and the ability to hold the prescribed orientation; it does not scale operator-commanded rotation. Here $\mathbf{D}$ denotes the controller's effective damping action rather than an independently identified physical Cartesian damping matrix.

The slave-controller API internally generated its normalized damping action from the active diagonal stiffness matrix and a scalar API parameter denoted by ζ. In the implementation convention,

$$\mathbf{D}_{\mathrm{API}}(c) = 2\zeta(c)\sqrt{\mathbf{K}(c)}, \tag{5}$$

where the square root is applied elementwise to the diagonal entries. $\mathbf{D}_{\mathrm{API}}$ denotes the API's normalized internal construction and is not reported as a physical damping matrix in SI units. Accordingly, ζ is treated only as the dimensionless damping-scaling parameter exposed by the controller API and is not interpreted as a Cartesian critical-damping ratio, because no effective Cartesian mass or inertia matrix was identified.

2.2.3. Master-side haptic rendering

For haptic rendering, the Franka internal state estimator provided the slave-side external-force estimate $\mathbf{F}_{\mathrm{ext}}$ in newtons. The baseline master-side force command along direction i was

$$u_{h,i}^{\mathrm{base}} = \mathrm{sgn}\left(K_f F_{\mathrm{ext},i}\right)\max\left(\left|K_f F_{\mathrm{ext},i}\right| - d, 0\right), \quad i \in \{x, y, z\}, \tag{6}$$

where K_f is a dimensionless force-scaling factor and d is a per-axis dead-zone threshold in newtons.

A separate positive-z cue was generated solely from the Omega.7 gripper angle θ_Ω:

$$\alpha_g = \mathrm{clip}\left(\frac{\theta_{\max} - \theta_\Omega}{\theta_{\max} - \theta_{\min}}, 0, 1\right), \tag{7}$$

$$u_g = \min\left(\alpha_g G_{\mathrm{cue}} F_{\mathrm{cue,max}}, F_{\mathrm{cue,max}}\right), \qquad u_{h,z} = u_{h,z}^{\mathrm{base}} + u_g. \tag{8}$$

With $G_{\mathrm{cue}} = 0.3$ and $F_{\mathrm{cue,max}} = 1.0$ N, the cue was $u_g = 0.3\alpha_g$ N and therefore remained within $[0, 0.3]$ N. It was updated during every nominal 200-Hz supervisory cycle, with a more open Omega.7 gripper producing a larger cue. This cue was derived solely from the Omega.7 gripper aperture and did not encode slave-gripper aperture or measured grasp force.

2.2.4. Gripper-execution channel

For gripper execution, v_g was passed as the `speed` argument of the `grasp` API, whose remaining arguments were `width`, `force`, `epsilon_inner`, and `epsilon_outer`; it was also used by `move` during opening and non-grasp position adjustments. The strategy value F_g specified the requested `force` argument of `grasp()` in newtons; it was not an independently measured fingertip force. The target grasp width was 0 m. In the experimental implementation, `epsilon_inner` was 0.005 m and `epsilon_outer` was 0.080 m (the maximum gripper aperture); these fixed tolerances defined the inner and outer width bounds used by the `grasp()` success criterion and did not vary by strategy. After contact and successful entry into the `HOLDING` state, the grasp command was not repeatedly retransmitted; the gripper maintained the grasp internally until `stop()` and the subsequent release or opening command.

The three channels act on distinct parts of the teleoperation system: Panda impedance determines the slave-side response to pose error and disturbance, Omega.7 settings render the estimated external force to the operator, and Franka Hand settings govern grasp execution. The proposed framework coordinates them through a common strategy-level parameter bundle.

2.3. Object-Conditioned Joint Parameter Configuration

2.3.1. Class-to-strategy and strategy-to-parameter mapping

The framework uses a three-level mapping from detected object class to an operational strategy and then to a joint parameter vector,

$$\Theta(c) = \{K_t, K_r, \zeta, K_f, d, v_g, F_g\}. \tag{9}$$

The strategy classes represent combined grasping risks rather than material softness alone. The apple and banana were assigned to the fragility-priority strategy because lower contact impact, closing speed, and requested grasp force were preferred to limit squeezing and deformation, while their smooth surfaces still posed a risk of slip. The paper cup and bottle were assigned to the balanced strategy. Excessive force can deform a paper cup, whereas overly low settings may produce an insecure grasp; the bottle similarly requires a compromise between gentle handling and slip resistance. The mouse and scissors were assigned to the stability-priority strategy because their smooth or irregular surfaces and transfer demands—and, for the scissors, elongated geometry—placed greater emphasis on pose retention and grasp stability. These assignments were specific to the present objects, platform, and task; the associated parameter settings are listed in Table 2.

2.3.2. One-shot triggering, locking, and parameter transition

At run time, the first threshold-qualified result read by the supervisory loop triggered a one-time update of the parameters enabled in the active mode, after which the strategy remained locked for the rest of the trial. Translational stiffness, rotational stiffness, and the damping-scaling parameter followed a smoothstep interpolation over approximately $T_s = 0.30$ s. With $\tau = \mathrm{clip}((t - t_0)/T_s, 0, 1)$ and $s(\tau) = 3\tau^2 - 2\tau^3$, each interpolated quantity $q \in \{K_t, K_r, \zeta\}$ was updated as

$$q(t) = q_0 + s(\tau)(q_{\mathrm{target}} - q_0). \tag{10}$$

The smoothstep interpolation yields continuous trajectories for K_t, K_r, and ζ, with zero update rate at the start and end of the transition. It was introduced to reduce discontinuities in the commanded impedance parameters and is not interpreted as a formal transient-stability guarantee for the complete teleoperation system.

The controller recomputed $\mathbf{D}_{\mathrm{API}}$ from the current $\mathbf{K}$ and ζ during this transition. Enabled values of K_f, d, v_g, and F_g changed immediately at the strategy event.

The class-to-strategy mapping is manually engineered. It converts a recognized known-object class into a constrained, repeatable operating strategy spanning the three subsystems; it does not infer physical properties directly. A new object must first be assigned a stable detector label and an explicit strategy entry before it can invoke a parameter bundle.

Table 2: Parameter settings for the three operational strategies. K_t is in N/m, K_r in N m/rad, d in N, v_g in m/s, and F_g is the requested `grasp()` force in N. ζ and K_f are dimensionless controller/API scaling parameters.

Strategy	Assigned objects	Slave impedance	Haptic interface	Gripper
Fragility-priority	Apple, banana	K_t = 50 N/m; K_r = 5 N m/rad; ζ = 0.8	K_f = 0.2; d = 0.3 N	v_g = 0.02 m/s; F_g = 8 N
Balanced	Paper cup, bottle	K_t = 150 N/m; K_r = 10 N m/rad; ζ = 1.0	K_f = 0.5; d = 0.4 N	v_g = 0.05 m/s; F_g = 15 N
Stability-priority	Mouse, scissors	K_t = 200 N/m; K_r = 13 N m/rad; ζ = 1.2	K_f = 0.7; d = 0.5 N	v_g = 0.10 m/s; F_g = 20 N

Figure 3 summarizes the class-to-strategy mapping, the one-shot configuration event, and the three parameter channels.

2.4. Parameter Design and Initial State

The fixed baseline vector was

$$\Theta_0 = (150,\ 10,\ 1.0,\ 0.5,\ 0.3,\ 0.10,\ 20), \tag{11}$$

with the same element order as $\Theta(c)$. The three strategy-level parameter sets were fixed before the formal experiment and were retained as the initial state whenever no automatic update occurred.

2.4.1. Engineering rationale for operating points

The three engineering-selected operating points were qualitatively screened on representative objects before formal data collection. Settings associated with sustained oscillation, noticeable haptic discomfort, grasp failure, or visible damage were excluded. The final settings in Table 2 were fixed before the 135 trials and should not be interpreted as optimized values or material-specific damage limits; the design rationale is reported in Supplementary Table S4.

2.5. Runtime Procedure

Algorithm 1: Object-conditioned joint-parameter scheduling

1. Load the initialization specified by the active experimental mode, start the nominal 200-Hz supervisory update, and launch the visual process only in modes C–E.
2. Poll the visual-result queue without blocking while executing teleoperation, haptic rendering, and gripper control.
3. While the strategy is unlocked, retain the initialized state when no result is available or confidence is below threshold. For a threshold-qualified result, select the mapped strategy; use the balanced strategy when the detected class is not explicitly mapped.
4. Update only the parameter channels enabled by the active mode. Interpolate K_t, K_r, and ζ over approximately 300 ms; update enabled haptic-interface and gripper settings immediately.
5. Lock the selected strategy for the remainder of the trial and ignore subsequent visual results for parameter configuration.
6. Record the active parameter state, task trajectory, duration, and outcome, and reset the system after the prescribed terminal procedure.

2.6. Bounded Fallback and Safety Measures

A misclassification as another known class invoked the strategy assigned to that class. A detected label without an explicit mapping invoked the balanced default strategy, whereas a missing or low-confidence detection caused no update. Every fallback outcome was confined to one of the three predefined parameter sets, although a bounded parameter set may still be inappropriate for a misclassified object.

In addition to these software fallbacks, robot collision detection, Franka Hand force settings, zero-force commands on exit, a standardized initial pose, and a manual emergency stop provided operational safeguards. The Omega.7 was operated using its standard driver and hardware configuration. The implemented mapping limited the additional master-gripper-aperture cue to 0.3 N. The baseline force-feedback command was transmitted without an additional software saturation layer, and saturation status was not logged.

3. Experimental Design

3.1. Experimental Objectives and Conditions

The experiment was designed to evaluate the system-level effect of object-conditioned joint parameter configuration during heterogeneous-object haptic teleoperation. Task completion time was the primary endpoint. The primary planned comparison was between the complete joint parameter-bundle mode (C) and the impedance-only reconfiguration mode (E). These modes shared the same visual-semantic display and slave-side impedance update; mode C additionally updated the haptic-interface and gripper-execution parameter groups. Thus, the C–E comparison tests the system-level contribution of the complete bundle beyond impedance-only reconfiguration, but does not identify the independent effects of the haptic and gripper groups or an interaction between them.

Comparisons of mode C with the fixed-parameter baseline (A), the manual-selection workflow baseline (B), and the visual-semantic display with fixed parameters (D) were specified as secondary planned contrasts. Success rate and Raw NASA-TLX were secondary descriptive outcomes. Master-side trajectory length and pause count were exploratory process measures. All analyses were interpreted as repeated-measures evidence within the tested three-operator sample rather than as population-level human-factors inference.

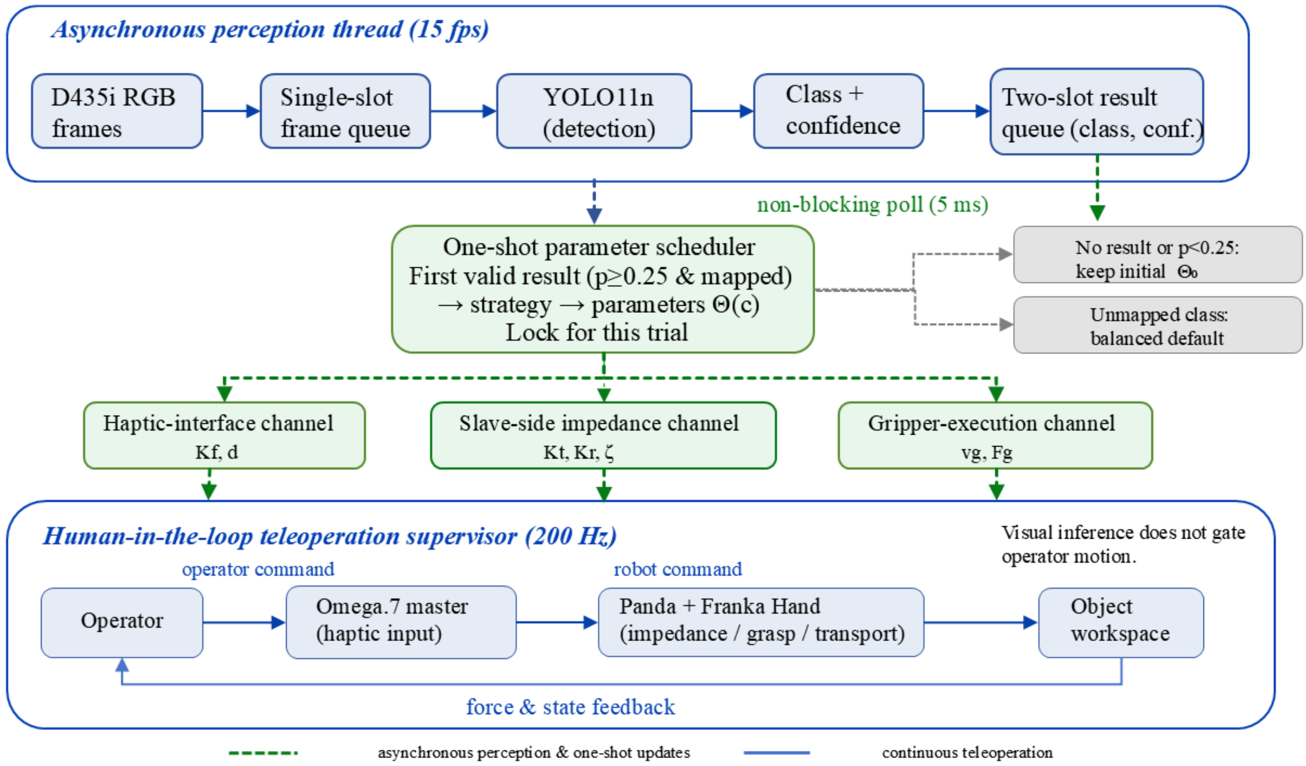

Figure 3: Object-conditioned joint parameter-configuration architecture. D435i RGB frames are processed asynchronously by YOLO11n and returned as class–confidence results through bounded queues. The 200-Hz teleoperation supervisor polls the result queue without blocking. The first mapped result with confidence ≥ 0.25 is converted into one strategy event $\Theta(c)$, which assigns the haptic-interface, slave-side impedance, and gripper-execution parameter groups and is locked for the trial. The diagram also shows the retained initial state for no or low-confidence results, the balanced default for an unmapped class, and the continuous operator–robot force/state-feedback loop.

Table 3 defines the five experimental modes, including the visual information available to the operator, the parameter groups that could change, and the role of each mode in the comparison. The visual-semantic display remained active throughout formal trials in modes C–E. Modes C and D held the visual-semantic display constant while differing in joint parameter configuration. Mode B required manual object identification and button-based strategy selection after timing had begun; its comparison with mode C consequently includes workflow automation and interaction differences and is not interpreted as an isolated controller effect.

3.2. Operators, Objects, and Matched Block Design

Three operators completed the main experiment (P01–P03; 23–24 years old; two male and one female; all right-handed). The Omega.7 was positioned on the left side of the platform and was operated with the left hand in every trial. This arrangement was held constant across operators and experimental modes to maintain a common physical interface condition. All operators had received basic teleoperation training and completed a 10–15 min familiarization session immediately before formal data collection to practice master-device motion, gripper input, and the task procedure. All operators provided written informed consent.

The six objects were selected to span different contact, grasping, and transfer demands, including deformability, smooth or irregular surfaces, and elongated geometry. They were assigned to the fragility-priority, balanced, or stability-priority strategy solely to select the parameter settings used in this platform and task. These assignments do not constitute a general classification of object material properties. Table 4 lists the physical profiles, assigned strategies, and principal manipulation risks; the contrasting paper-cup and scissors cases illustrate the balance between deformation avoidance and orientation retention.

The experiment comprised 27 matched task blocks. A block was defined by a fixed operator, object, and repetition index and contained one trial in each of modes A–E. Accordingly, 27 blocks $\times$ 5 modes per block yielded 135 trials. This structure makes the five mode observations within each block the pairing unit for the completion-time analysis.

Within each operator and strategy category, the two objects were assigned to the three repetitions in a fixed 2:1 ratio. The allocation was held constant across the five modes within a block, preserving the object-matched comparison, and the 2:1 allocation was reversed across operators. The resulting object counts gave the approximate 5:4 balance within each strategy shown in Table 5. This allocation was specified by the experimental schedule, rather than selected after observing performance.

Five-mode orders were preassigned at the operator-by-strategy-group level using nine different sequences, so the design did not rely on a single fixed order. The schedule dis-

Table 3: Experimental modes, channel-update logic, and comparison roles.

Mode	Visual cue / selection	Slave impedance	Haptic interface	Gripper execution	Comparison role
A: Fixed-parameter baseline	None	—	—	—	Fixed-parameter baseline.
B: Manual-selection workflow baseline	No visual cue; operator selects a complete strategy after timing starts.	Manual	Manual	Manual	Workflow baseline; timing includes manual identification and selection.
C: Joint parameter-bundle configuration	Class, strategy, confidence, and colored bounding box	Auto	Auto	Auto	Full joint configuration.
D: Visual-semantic display with fixed parameters	Same as C	—	—	—	Visual-semantic display control.
E: Impedance-only reconfiguration	Same as C	Auto	—	—	Impedance-only control.

Note: Auto denotes a one-shot update after the first valid visual detection; Manual denotes manual strategy selection after timing starts; — retains the initialized setting. In modes C–E, no detection or confidence below 0.25 retains the initialized state. A detected class without an explicit mapping invokes the balanced strategy; in mode D, the visual-semantic output is displayed while Θ_0 is retained. Vision is not used in modes A and B. The C–E contrast changes the haptic-interface and gripper parameter groups together and therefore does not identify their individual effects or an interaction between them.

Table 4: Test objects, assigned strategies, and principal task risks.

Object	Assigned strategy	Physical profile	Principal task risk
Apple	Fragility-priority	~200 g; smooth; ∅70–80 mm	Impact or slip; gentle contact required.
Banana	Fragility-priority	~120 g; smooth; 20 × 180 mm	Compression-induced deformation and slip.
Paper cup	Balanced	~5 g; paper; ∅75×90 mm	Easily deformed; stable grasp required.
Bottle	Balanced	~30 g; smooth plastic; ∅65×200 mm	Slip; balance between efficiency and stability.
Mouse	Stability-priority	~100 g; smooth plastic; 65 × 120 × 35 mm	Rigid irregular surface; transfer slip.
Scissors	Stability-priority	~150 g; metal and plastic; 50 × 170 × 15 mm	Rigid elongated geometry; high orientation demand.

Table 5: Allocation of matched task blocks and trials by strategy and object.

Object	Blocks	Trials
Fragility-priority		
Apple	4	20
Banana	5	25
Balanced		
Paper cup	5	25
Bottle	4	20
Stability-priority		
Mouse	5	25
Scissors	4	20
Total (six objects)	27	135

Note: Each block contains one trial in each of the five modes.

tributed modes across early and late positions but was not a complete counterbalanced Latin-square design. Across the nine sequences, mode C occurred in position four in three groups, whereas mode E occurred in position five in three groups. The exact sequences are reported in Supplementary Table S2. With three operators, order, object, and operator effects could not be fully separated; C–E is therefore interpreted as a within-sample paired contrast rather than an order-free population effect.

3.3. Task Procedure and Outcome Measures

Before each trial, the test object was manually placed within a predefined work region and at an approximately prescribed orientation. Small reset errors in position and orientation were unavoidable, but the same placement protocol was used for all modes. Only translational master motion was mapped; the desired Panda end-effector orientation remained fixed throughout the trial.

Each trial followed reset, approach, grasp, transfer, release, and termination stages. In modes C–E, camera acquisition, visual inference, and teleoperation started concurrently at task initiation; visual output did not gate operator motion. In mode B, timing began before object identification and strategy selection, and robot motion was permitted only after the operator completed the button selection. No separate timestamp marked the end of this manual selection step.

A trial was successful when grasping, transfer, and placement were completed without a drop, observable slip, or visible damage. Complete separation of the object from the gripper was classified as a drop; visible relative motion between object and gripper during grasp or transfer as slip; and a clear indentation, crushing deformation, or other visible damage as damage. A designated researcher applied these predefined criteria in real time. Trials were not systematically video-recorded, and outcome labels were not independently or blindly re-evaluated.

Failure did not stop the timer immediately. Regrasping after a drop was not permitted; the failure was recorded, and the operator completed the prescribed return-and-termination procedure until the robot and gripper reached the predefined terminal state. The same timing endpoint was therefore applied to successful and failed trials. The system recorded master-side trajectory, gripper input, active control parameters, task duration, and outcome. It did not retain event-level visual classes, confidences, strategy-trigger timestamps, or synchronized trigger–contact timestamps; consequently, the retained record cannot retrospectively verify pre-contact strategy selection in every trial.

Task completion time was measured from task initiation to the common terminal state described above and was retained for successful and failed trials. Success rate was calculated as the number of successful trials divided by all trials in a mode and is reported descriptively because outcome labels were assigned in real time without independent review.

Subjective workload was assessed with Raw NASA-TLX [25]. Each of the six dimensions was rated from 0 to 100, and their unweighted arithmetic mean was reported. Each operator completed one questionnaire immediately after the three trials in a given strategy × mode condition. These trials combined the two objects assigned to that strategy in the operator-specific 2:1 ratio. The workload dataset therefore contained 3 operators ×3 strategies ×5 modes = 45 questionnaires, or nine questionnaire units per mode. It supports mode- and strategy-level descriptive comparisons, but not object-specific or population-level workload inference.

Master-side trajectory length and pause count were treated as exploratory process measures. Visual performance was characterized separately in a controlled image test using class accuracy, class-to-strategy mapping accuracy, confidence, and per-image processing time. A pause was operationally defined before analysis as a continuous interval in which master-device speed was below 0.005 m/s for at least 0.30 s. Pauses were obtained by differentiating raw master-side trajectories sampled at approximately 200 Hz.

3.4. Statistical Analysis

Completion time across the five modes was first assessed with a Friedman test using the 27 matched task blocks as the repeated-measures units. When the omnibus test was significant, paired Wilcoxon signed-rank tests with Holm–Bonferroni correction were used for the planned contrasts. For the primary C–E contrast, the analysis reports the paired mean difference, relative change, and a 95% bootstrap confidence interval based on 10,000 resamples of matched task blocks. Effect direction was also summarized by operator, together with a leave-one-operator-out sensitivity analysis, to assess whether a single operator dominated the observed difference. Because blocks were nested within operators, these analyses characterize conditional repeated-measures evidence in the observed sample rather than independent population-level observations.

Paired C–E success outcomes were tabulated, and an exact McNemar test was reported only as a supplementary analysis because few discordant pairs were observed. Raw NASA-TLX was compared descriptively across the nine operator × strategy questionnaire units per mode; no bootstrap interval was calculated. Trajectory length and pause count were analyzed as exploratory measures. Results are reported, as appropriate, as median [IQR] and mean ± SD. Because the 27 blocks are nested within only three operators, all inferential results and bootstrap intervals are conditional on the present repeated-measures sample and do not support population-level inference across operators.

4. Results

4.1. Controlled Visual Test and Asynchronous Integration

The separate closed-set visual test comprised 180 photographs (30 per class) acquired under fixed viewpoint, background, and lighting conditions. All images were classified correctly and mapped to the intended strategy; mean confidence was 0.853 and mean per-image inference time was 48.19 ms (Fig. 4; class-wise results and the confusion matrix are provided in Supplementary Table S5 and Fig. S1). These results characterize the six known object instances under controlled conditions and do not assess online trigger correctness, trigger-to-contact timing, or performance on untested objects.

The median logged supervisory-cycle duration was 5.07 ms.

4.2. Results Across the Five Modes

Across the five modes, C had the lowest median completion time, the highest observed success rate, the lowest Raw NASA-TLX score, and the lowest median master-side trajectory length (Table 6 and Figs. 5 and 6). Using the trial-level values in Supplementary Table S1, mean completion time in mode C (19.28 ± 1.30 s) was lower than in modes A (21.42 ± 1.58 s), B (21.01 ± 1.61 s), D (20.91 ± 1.10 s), and E (21.07 ± 1.56 s) by 10.0%, 8.2%, 7.8%, and 8.5%, respectively. As prespecified, inferential analyses focused on completion time as the primary endpoint.

Treating the 27 operator-nested matched task blocks as within-sample repeated-measures units, the Friedman test indicated differences in completion time among the five modes ($\chi^2(4) = 30.904$, $p < 0.001$). Holm-adjusted paired Wilcoxon tests showed lower completion times in mode C than in modes A, B, D, and E (all $p < 0.01$). These p values provide conditional evidence for the observed sample and are not population-level participant inference. Descriptively, mode C also had the lowest Raw NASA-TLX score in all nine operator × strategy units.

4.3. Primary C–E Comparison and Exploratory Process Measures

The primary C–E comparison is summarized in Table 7 and Fig. 7.

The median completion times were 19.57 s in mode C and 20.73 s in mode E. With $\Delta T = T_E - T_C$, the mean paired difference was 1.79 s (matched-block bootstrap 95% CI, [1.10, 2.51] s), corresponding to a mean reduction of 8.5%. Completion time was lower in mode C in 22 of the 27 matched blocks. The mean differences for P01, P02, and P03 were 1.66, 2.56, and 1.16 s, respectively. Across the six objects, the relative differences ranged from 3.3% to 13.2% (Fig. 8).

As a post hoc sensitivity analysis, the C–E completion-time comparison was repeated after retaining only the 23 matched blocks in which both trials were successful. The mean paired difference was 1.80 s, and 19 of the 23 blocks favored mode C; this estimate was similar in magnitude to the primary analysis.

For paired success outcomes, 23 blocks succeeded in both modes, three succeeded in mode C and failed in mode E, one failed in mode C and succeeded in mode E, and none failed

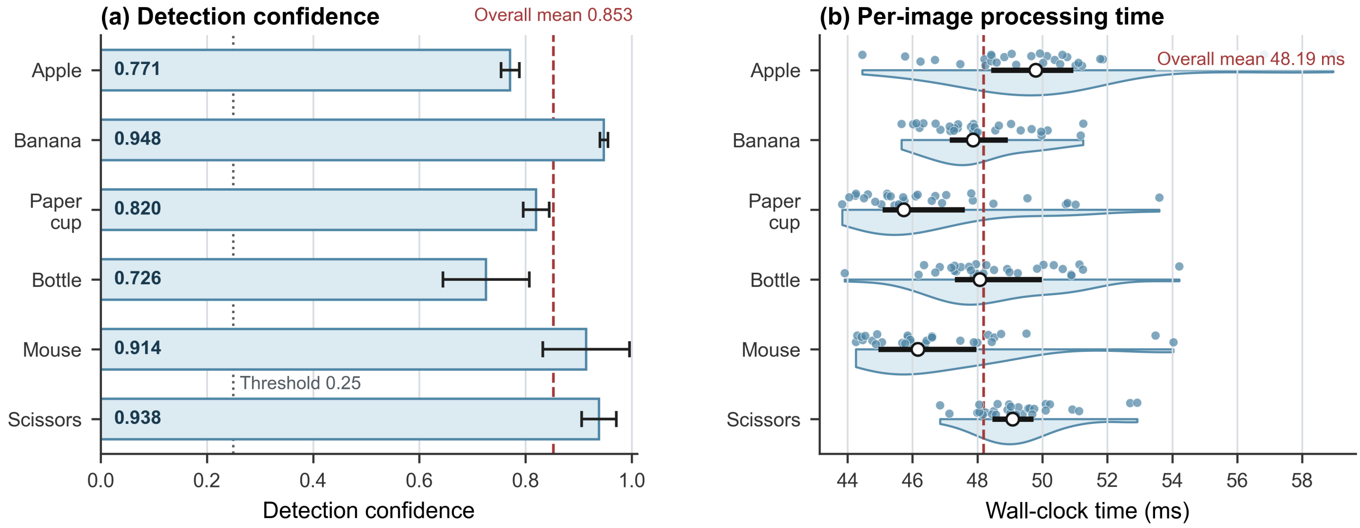

Figure 4: Confidence and per-image inference time in the closed-set visual test (n = 30 per class). Bars show mean ± SD; half violins, dots, segments, and open circles show distributions, observations, IQRs, and medians. Reference lines mark the 0.25 threshold and overall means (0.853; 48.19 ms). Timing is not end-to-end task-trigger latency.

Table 6: Descriptive results across the five experimental modes.

Mode	Completion time (s)	Trajectory length (m)	Success rate	Raw NASA-TLX
A	21.18 [20.62, 22.08]	0.757 [0.693, 0.816]	22/27 (81.5%)	62.50 [59.67, 64.50]
B	20.89 [20.12, 21.83]	0.787 [0.721, 0.861]	21/27 (77.8%)	56.17 [55.00, 59.33]
C	**19.57 [18.41, 20.05]**	**0.697 [0.660, 0.769]**	**26/27 (96.3%)**	**48.67 [47.67, 51.83]**
D	20.79 [20.32, 21.16]	0.722 [0.678, 0.768]	24/27 (88.9%)	60.33 [57.33, 62.50]
E	20.73 [19.95, 22.25]	0.732 [0.678, 0.799]	24/27 (88.9%)	53.67 [51.83, 57.83]

Note: Values are median [IQR], except success rate. Completion time, trajectory length, and success rate use n = 27 trials per mode; Raw NASA-TLX uses n = 9 strategy-level questionnaire units per mode. Mode definitions are given in Table 3; trial-level performance data are provided in Supplementary Table S1.

Table 7: Within-sample primary comparison of the complete parameter bundle and impedance-only reconfiguration.

Measure	C, median [IQR]	E, median [IQR]	Mean Δ (E–C), 95% CI	Operator direction
Completion time (s)	19.57 [18.41, 20.05]	20.73 [19.95, 22.25]	1.79 [1.10, 2.51]	3/3 favored C
Trajectory length (m)	0.697 [0.660, 0.769]	0.732 [0.678, 0.799]	0.024 [-0.014, 0.059]	Mixed
Raw NASA-TLX	48.67 [47.67, 51.83]	53.67 [51.83, 57.83]	4.87 [not calculated]	3/3 favored C

Note: Positive Δ values favor mode C. Completion time and trajectory length use n = 27 matched task blocks; Raw NASA-TLX uses n = 9 questionnaire units per mode. Intervals are matched-task-block bootstrap 95% CIs from 10,000 resamples and quantify uncertainty across the observed blocks, not a population-level participant effect; no bootstrap interval was calculated for Raw NASA-TLX.

in both modes. With only four discordant pairs, the exact two-sided McNemar test did not establish a difference (p = 0.625). The observed success-rate difference is therefore interpreted as supplementary descriptive evidence rather than as a confirmed mode effect.

The median master-side trajectory lengths were 0.697 m in mode C and 0.732 m in mode E. The mean paired difference was 0.024 m, with a bootstrap 95% CI of [-0.014, 0.059] m; the interval included zero. Pause counts were lower on average in mode C but had the same median in modes C and E.

For the exploratory pause measure, using the prespecified definition in Section 3.3, mode C had a median pause count of 3 [2, 3.5] and a mean of 2.74 ± 1.23, whereas mode E had a median of 3 and a mean of 3.41 ± 1.67. Mode C had a lower mean in each strategy class, but the two modes had the same overall median. No formal hypothesis test was prespecified or performed for pause count, and no threshold-sensitivity analysis was available.

4.4. Failure Analysis

Eighteen failures occurred across the 135 trials: 5, 6, 1, 3, and 3 in modes A–E, respectively. The failures included object drops, observable slip, and visible damage, as summarized in Table 8. One designated researcher recorded outcomes in real

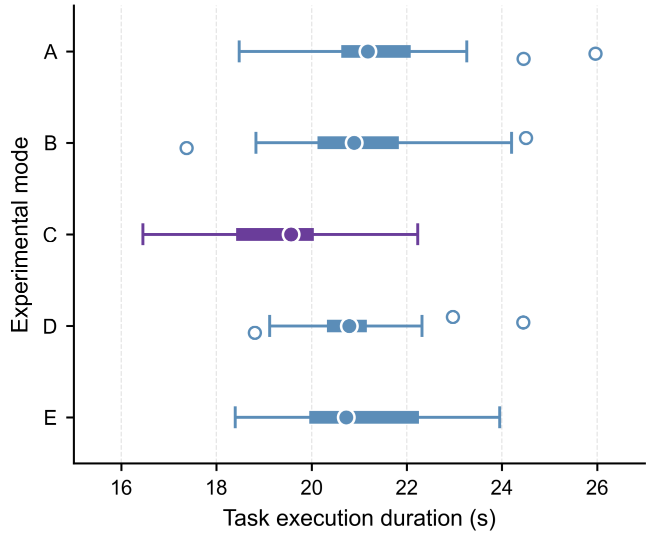

Figure 5: Task duration across the five modes (n = 27 matched blocks per mode). Filled circles, thick segments, whiskers, and open circles show medians, IQRs, 1.5 × IQR ranges, and outliers. This descriptive plot does not show block pairing; the primary C–E pairing is shown in Fig. 7.

time using the predefined criteria in Section 3.3; no systematic video recording or independent re-rating was available.

Mode C had one failure in 27 trials, the lowest observed count among the five modes under the present experimental conditions.

4.5. Consistency Across Operators and Objects

Figure 8 reports paired C–E differences by operator and object. At the operator level, the mean paired differences $\Delta T = T_E - T_C$ for P01, P02, and P03 were 1.66, 2.56, and 1.16 s, respectively. The corresponding median [IQR] differences were 1.91 [0.60, 3.14], 2.63 [2.19, 3.34], and 1.60 [-0.53, 1.73] s. Mode C was faster in 7/9, 9/9, and 6/9 task blocks, respectively. After P01, P02, and P03 were excluded in turn, the mean differences in the remaining blocks were 1.86, 1.41, and 2.11 s, respectively; all continued to favor mode C.

At the object level, mode C had the shortest mean completion time among the five modes for each of the six objects (Supplementary Table S6). Relative to mode E, the reductions ranged from 3.3% for the bottle to 13.2% for the scissors; the direction was consistent across objects, although the magnitude varied.

5. Discussion

5.1. Principal Findings and Interpretation of the Five-Mode Design

The five-mode evaluation yielded three main observations. Mode C had the shortest completion time among the tested modes. In the prespecified C–E comparison, the complete parameter bundle reduced mean completion time by 1.79 s (8.5%), with a matched-block bootstrap 95% CI of 1.10–2.51 s; the direction favored mode C for all three operators and all six objects. Success rate and Raw NASA-TLX descriptively favored mode C, whereas the trajectory-length interval included zero. Taken together, these findings support a within-sample system-level advantage of the complete joint configuration under the tested conditions, rather than an isolated effect of any one parameter channel.

The five modes clarify the scope of this interpretation. Mode B included object identification and manual strategy selection after timing began, so its contrast with mode C represents an automated-workflow comparison rather than an isolated controller comparison. Modes C–E retained the same visual-semantic display throughout the trial. In particular, mode D displayed the same class, mapped strategy, confidence, and bounding box as mode C while retaining fixed parameters; the C–D pattern is therefore consistent with an additional task-level benefit beyond displaying the semantic cue alone. The primary C–E contrast is more specific: both modes used the same class-to-strategy mapping, visual-semantic display, and slave-side impedance update, while mode C additionally updated the master-side haptic gain, force dead zone, gripper speed, and requested grasp-force setting.

Accordingly, the C–E result is consistent with an additional system-level benefit from configuring the haptic-interface and gripper settings together with slave-side impedance. It does not identify the contribution of either added channel or test an interaction between them. The completion-time difference was not accompanied by clear evidence of a shorter master-side geometric path, and the lower mean pause count in mode C remained exploratory because the two modes had the same median and no confirmatory pause analysis was prespecified. The available process measures may be compatible with smoother task progression or fewer corrective interruptions, but they do not establish a specific mechanism.

5.2. Asynchronous Perception–Control Integration

A one-shot configuration event does not require semantic perception to operate at the supervisory-loop rate. In the present implementation, 15-fps image acquisition and independent inference exchanged data with the nominal 200-Hz supervisory update through bounded queues. The single-slot frame queue limited stale-image accumulation, and non-blocking polling prevented the supervisory loop from waiting for visual output. This architecture is suited to task-initial configuration rather than continuous visual servoing.

The implementation therefore demonstrates asynchronous, non-blocking integration of low-rate semantic perception into the software architecture. It does not establish hard real-time performance, a measured end-to-end trigger latency, or completion of configuration before contact in every trial.

5.3. Variation Across Operators and Objects

The mean paired C–E difference was positive for all three operators and remained positive when each operator was excluded in turn. Across the six objects, mode C was faster than mode E, with descriptive mean reductions from 3.3% to 13.2%. These directional checks indicate that the observed aggregate

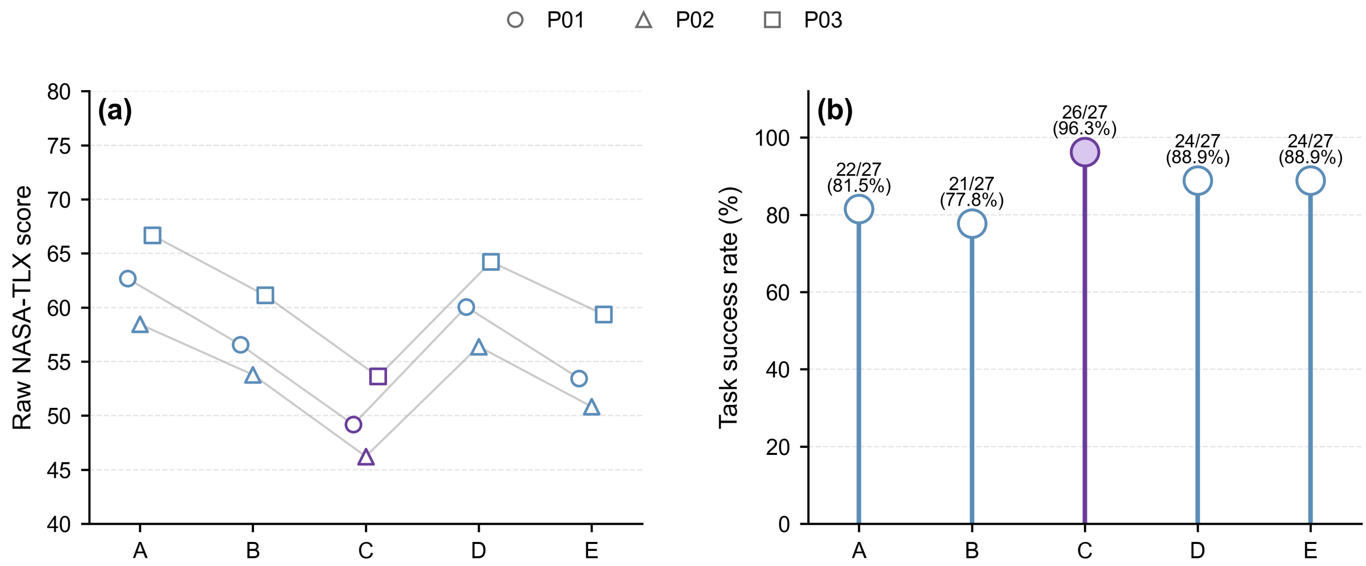

Figure 6: Human workload and observed task success across the five experimental modes. (a) Raw NASA-TLX scores: connected markers show each operator's mean across three strategy-level questionnaire units per mode (n = 9 questionnaire units per mode); marker shape identifies the operator. (b) Observed successful trials out of 27 for each mode. These panels are descriptive and do not display confidence intervals or inferential tests.

Table 8: Observed failures and representative observations by experimental mode.

Mode	Failures/total	Representative observations
A	5/27	Paper-cup deformation and unstable positioning of the scissors.
B	6/27	The available real-time records did not support attribution of a specific failure cause.
C	**1/27**	Mouse slip during transfer.
D	3/27	Object drops or loss of grasp during holding.
E	3/27	Observed drops, slip, or visible damage during grasp establishment or transfer.

Note: Mode definitions are given in Table 3. Observations are descriptive and were recorded in real time.

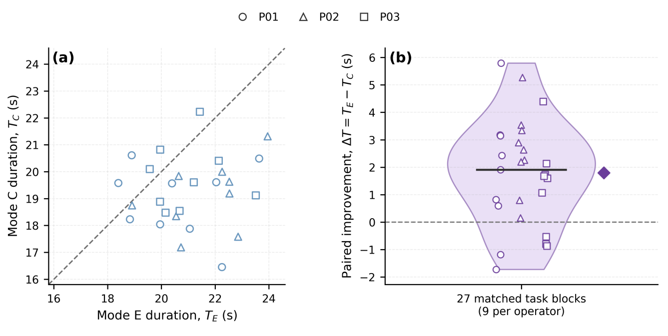

Figure 7: Paired C–E completion time across 27 matched blocks. Points below the identity line and positive $\Delta T = T_E - T_C$ favor C. The violin, segment, and diamond show the distribution, median, and mean. Mean ΔT was 1.79 s (matched-block bootstrap 95% CI, 1.10–2.51 s); 22/27 blocks favored C.

difference was not visibly driven by a single operator or object in the present dataset; they do not remove the operator nesting or support population-level inference.

The magnitude varied across objects: it was smaller for the bottle and banana and larger for the paper cup and scissors. Such variation may reflect differences in grasp pose, deformation risk, and orientation-stability requirements. However, each object contributed only four or five matched blocks; the object-level results remain descriptive and cannot establish an object-specific mechanism or generalization to untested objects.

5.4. Relation to Previous Work and Mechatronic Design Implications

Previous tele-impedance and variable-impedance approaches commonly adapt remote impedance according to human state, contact force, demonstrations, or task phase [6–12, 26]. Visual methods have translated object attributes, geometry, and vision–language inputs into stiffness or impedance matrices [16, 17, 19, 20], while automatic switching, shared control, and multi-stiffness interfaces have addressed mode selection and control-authority allocation [18, 21, 22, 27, 28]. Teleoperated grasping has also used tactile deformation and grip-safety information to regulate gripper impedance [23].

The present work complements these approaches with a constrained systems architecture in which one semantic strategy event assigns a consistent operating state across slave mechanical response, operator-facing haptic rendering, and gripper execution. The contribution is thus a cross-subsystem, task-level

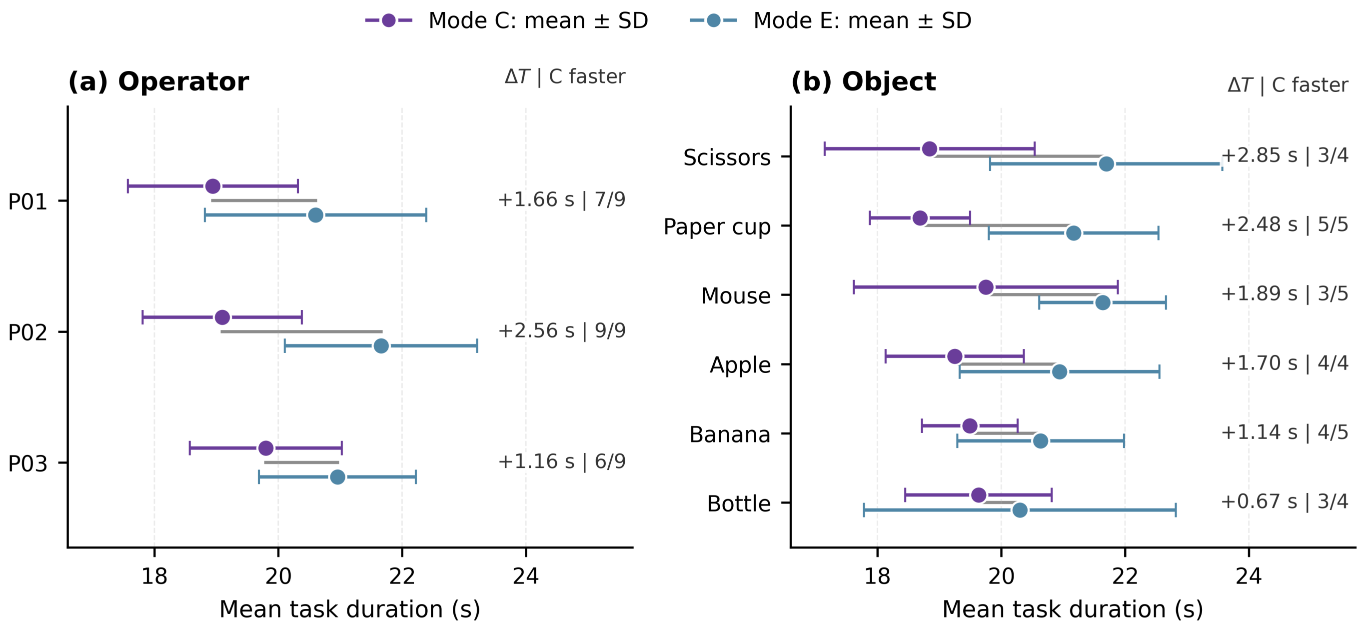

Figure 8: Mean C–E task duration by operator and object. Points and error bars show mean ± SD; connectors link paired means. Labels give $\Delta T = T_E - T_C$ and blocks favoring C. Object-level summaries are descriptive (n = 4 or 5 blocks).

configuration architecture and its system-level evaluation, rather than a new impedance-control law or a dynamically derived coupling law. Bounded queues, non-blocking polling, one-shot triggering, and strategy locking allow low-rate perception to remain outside the nominal high-rate supervisory path while avoiding repeated intra-trial reconfiguration.

This design principle may be useful in semi-structured applications such as flexible sorting, remote maintenance, and dismantling, where a limited set of recurring object or task classes can be associated with validated operating strategies. Scaling beyond a small class set would require sustained maintenance of the mapping and potentially more than three strategy levels. Additional tactile or contact-state sensing could support post-contact verification or correction without placing the visual process in the high-rate supervisory path.

5.5. Limitations and Future Work

Formal trials did not retain complete visual-event histories or synchronized strategy-event–contact timestamps. Trigger correctness, fallback use, trigger–contact intervals, and completion of configuration before contact therefore could not be reconstructed. The controlled 180-image test characterizes recognition only under fixed imaging conditions; the framework should be interpreted as designed for contact-preparatory configuration, not as having verified pre-contact configuration in every trial.

The experiment used repeated measurements from three operators, six known objects, translational master motion, fixed end-effector orientation, prescribed object placement, and pre-assigned sequences that were not fully counterbalanced. The manually engineered mapping requires an explicit detector label and strategy entry for each new object. The results are consequently limited to the present platform, task, and participant sample and do not support population-level human-factors inference.

The parameter sets were qualitatively screened rather than optimized, and the C–E contrast changed haptic-interface and gripper settings together; channel-specific ablations are therefore needed. Outcome assessment lacked video-based independent review, and the retained records did not provide direct contact-quality or corrective-action measures. Future work should add synchronized event and contact logging, channel-level ablations, direct contact measures, and broader operator and object samples.

6. Conclusion

This study presented an object-conditioned strategy-level parameter-bundle configuration framework for haptic teleoperation of heterogeneous objects. A semantic mapping assigned each detected object class to an interpretable operational strategy that jointly specified slave-side impedance, master-side haptic rendering, and gripper execution. An asynchronous, one-shot, strategy-locked update integrated low-rate visual perception without placing inference in the nominal 200-Hz supervisory call path.

Across 27 matched task blocks comprising 135 physical trials, the complete parameter bundle reduced mean completion time by 1.79 s relative to impedance-only reconfiguration, corresponding to an 8.5% reduction (matched-block bootstrap 95% CI, 1.10–2.51 s). The direction favored the complete bundle across all three operators and six objects. This mode also had the highest observed success rate and the lowest Raw NASA-TLX score, although these outcomes were interpreted descriptively.

No clear reduction in master-side trajectory length was established.

These findings provide within-sample, system-level evidence that object-conditioned configuration across the slave robot, haptic interface, and gripper can offer an additional task-performance benefit beyond impedance-only reconfiguration under the tested conditions. The present design does not isolate the individual contributions of the haptic and gripper channels or verify trial-level trigger–contact timing. Channel-specific ablations, synchronized event logging, direct contact measurements, and a larger participant sample are therefore needed. Within these boundaries, the framework provides an interpretable system-design approach for object-dependent configuration in semi-structured haptic teleoperation tasks.

Supplementary Material

Supplementary Tables S1–S6 and Fig. S1 are supplied as a separate supplementary file.

Declarations

Ethics statement

The school confirmed that formal ethics committee review was not required for this low-risk, non-invasive study. All participants provided written informed consent before participation. The experimental procedure was conducted in accordance with relevant institutional guidelines; no personally identifiable information was collected, and all data were anonymized.

Image integrity

Figure 1 is a photograph of the actual experimental platform. No generative AI was used to add, remove, replace, or reposition experimental elements. Figures 2–3 were redrawn by the authors using Python/Matplotlib, and Figures 4–8 were generated from experimental data. The authors verified the logic, text, and data mapping in the final figures.

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Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The reproducibility bundle prepared for this study contains de-identified trial-level completion time, master-side trajectory length, outcome data, raw NASA-TLX responses, closed-set vision-validation data, analysis scripts, final figure assets, and available figure-source files. Supplementary Tables S1–S6 and Fig. S1 are included in the bundle. Before submission, the authors will release a versioned GitHub repository and archive the corresponding release in Zenodo; this statement will then be replaced with the repository name, permanent link, version-specific DOI, and data citation. Event-level visual/strategy/contact logs, video, and detailed failure-type records were not retained and therefore cannot be retrospectively shared.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used OpenAI Codex to assist with language editing, consistency checks, and the drafting of Python/Matplotlib code. The authors subsequently reviewed and edited all outputs, verified the technical statements and data mappings, and take full responsibility for the content of the manuscript.

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